

Unearth

Eliminating Wasteful Spending for General Contractors

How to identify and get rid of
over \$1 million in unproductive
expenditures.



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I. The cost of wasted time.

Projections for 2018 predict that US construction companies will spend \$177.5 billion for work that doesn't move their projects toward completion.¹ Why? Because their teams don't have the tools they need to be effective, leading them to spend an average of 35% of their time dealing with distractions.²

Construction's productivity crisis is no secret, and this \$177.5 billion sum is just one piece of the \$1.6 trillion in unrealized productivity.³ However, it's one of the most critical pieces to focus on because it's immediately addressable and yields instant ROI.

The industry is stuck between two worlds: paper and digital. With the rise of the digital era, technology has inevitably crept into different construction processes; smartphones, the internet, and email are unavoidable parts of every project. The issue is that the construction industry never adapted to these developments—they've continued to use the old systems and processes created when paper was king.

Sitting in the middle of the digital and paper worlds is the primary reason construction spends \$177.5 billion on unproductive activities. Teams can either use paper processes or digital processes, not both. Trying to use both leaves you worse off by creating duplicate work: someone has to make your paperwork digital, inevitably causing an organizational disaster.

Going back to an all paper system is not an option; nobody will hire a contractor that relies on paper. So, the industry needs to adapt to a new set



In 2018, the amount US construction companies paid for unproductive activities totaled

\$177.5 billion

The global productivity gap for construction was

\$1.6 trillion

of digital tools and processes if they're going to stop wasting billions trying to hold onto old ways.

Fortunately, it's not as hard as it sounds. In this paper, we'll identify the types of activities that waste time, demonstrate how to calculate the value of this wasted time, show you how to fix the problems, provide you with some real life examples, and then guide you on the best ways to get started.

1. Thomas, E., & Bowman, J. (2018, August 1). *Construction Disconnected* (Publication). Retrieved October 1, 2018, from Plangrid & FMI website: <https://www.plangrid.com/ebook/construction-disconnected/>, 6.
2. *Construction Disconnected*, 6.
3. Barbosa, F., Woetzel, J., & Mischke, J. (n.d.). *Reinventing Construction: A Route to Higher Productivity* (Working paper). McKinsey&Company. Retrieved October 1, 2018, from <https://www.mckinsey.com/industries/capital-projects-and-infrastructure/our-insights/reinventing-construction-through-a-productivity-revolution>.

II. How going digital eliminates unproductive activities.

How do employees cost construction companies almost \$200 billion/year in unproductive work? To answer that question, you need to categorize the activities an employee can engage in on a project. Construction work typically fall into one of three buckets:

1. **Project-Focused**
2. **Client-Focused**
3. **Complication-Focused**

Things that fall under the project and the client are productive activities that are necessary to move the project forward. Things under complications are unproductive in the sense that the project either isn't actively moving forward or they're caused by an avoidable setback.

You want your team focused primarily on activities that move the project forward, while any client-focused work should be effective and concise. And, in a perfect world, you'd have zero work stemming from complications. Before this can be your reality, you have to understand what's going wrong.

Project-Focused

- Coordinating project teams
- Executing project deliverables

Client-Focused

- Stakeholder meetings
- Project reporting

Complication-Focused

- Tracking down data
- Rework
- Dispute resolution

1. The issue with **PAPER** in a digital world

Paper may be easy, but it is definitely not convenient. It's easily damaged or lost. It can't be sent instantly. Changes must be made on a computer, reprinted, and then redistributed. It's costly. People can't collaborate with it unless they're in the same room.

In a world where paper is the only option—these aren't roadblocks because everyone is on the same page. In a digital world, however, it's a recipe for disaster.

Expectations are different today. Everybody expects to be working off the most current information. When teams

have outdated information, they make mistakes. These mistakes cost time and money to be fixed, meaning one piece of outdated paper could be the difference between a profit and a loss.

Mixing digital assets with paper simply creates too much chance for error. Not to mention the tedium of transcribing paperwork to digital documentation. If you want to eliminate complications, and the wasted work that is associated with them, you cannot rely on paper in today's world. However, this brings up a second issue: organizing and managing digital data.

2. The issue with POORLY ORGANIZED DIGITAL DATA

Digital records suffer from the opposite problem of paper: they can be too easy and convenient. Creating digital data is cheap and simple, meaning employees can unintentionally cause a company to have data overload.

Too much data can be just as damaging as relying on paper processes. A glut of poorly organized data can make it extremely difficult to find what you need and update it.

Just consider photo documentation. We have customers coming to us with over 50k photos of a single project. Not only are they not organized, they're often spread across

different devices - some in the cloud, some on USB drives, and some just on people's devices. When a dispute arises they are paralyzed; no one has the time to sift through that many photos, especially if the evidence they need might not even be there!

Photos are just one type of data, extrapolate the problem out to include your RFIs, emails, drone flights, plans, as-builts, etc., and it's clear how quickly things get out of hand. You have to be digital, but you have to keep your digital data streamlined and organized or you won't be much better off than you were with paper alone.

These two problems comprise a larger dilemma faced by construction teams. The purpose of going digital isn't just to move your paper workflows to a computer.

The purpose of going digital is to run your business differently, enabling your teams to do more with less.

- **Equip teams to collaborate remotely on distant projects**
- **Reduce mistakes and eliminate rework by keeping all plans current**
- **Streamline planning and logistics with rich, mixed dataviz instead of hand waving**
- **Perform real-time QA & QC to catch issues at the start and resolve them instantly**

By creating a fresh digital workflow, your team can concentrate on priority #1: project-focused work, while keeping customer-focused tasks concise and eliminating complications. Seamless data organization means team can simply access the data when they need it, stakeholders stay up-to-date without inconvenient meetings, communications are missed less often, and everyone stays working off the most current data set.

Even if you understand how your data management is causing lost time, you might not realize how much money it's ultimately costing you. In the next section of this paper, we'll take a look at how you can determine the cost your firm.



III. How does wasted time cost millions of dollars and what is it costing me?

It can be easy to write-off unproductive activities caused by poor data management as a cost of doing business. But, do you know how much money you're actually leaving on the table? Wasted time costs millions of dollars because every minute an employee spends doing something unproductive quickly adds up when multiplied across hundreds of people.

Estimating your spending on unproductive time is relatively easy by looking at national averages and applying them to your current staff. Take a look at the table below for an idea of the time and cost associated with unproductive activities for different construction roles.⁴

So, for instance, if you're part of a small construction firm and you're curious what you could be spending on wasted time per month, you'd create a table similar to this one and add a column for number of positions (green columns).

That means that an average small construction firm in the United States spends \$76,675/month or close to \$1,000,000/year for your employees to do work that doesn't actively move projects forward. That's a conservative estimate. If a small firm is leaving over a million dollars per year on the table, it's easy to see how the entire US spends \$177.5 billion on wasted time.

Can you imagine how different your budget would look if your team spent those hours getting your projects across the finish line? Not to mention the amount of money that would be saved by eliminating mistakes and rework. With this in mind, take a few minutes to get an estimate of how much your firm is likely spending on this problem. Is it a number that's worth doing something about to you? The next sections of this paper will show you how this problem can be fixed and how you can begin to recover the millions of dollars going to waste.

Role	Hours Weekly	Hours Monthly	Monthly Cost	Annual Cost	Number of Positions	Monthly Cost to Your Company
C-Level	14.05	56.2	\$10,440.51	\$125,286.10	4	\$41,762.04
Directors/ VPs	13.52	54.08	\$5,855.02	\$70,260.26	2	\$11,710.04
PMs	13.39	53.56	\$4,103.94	\$49,247.33	3	\$12,311.82
Supers	15.84	63.36	\$4,068.78	\$48,825.32	2	\$8,137.56
Estimators	12.15	48.6	\$2,754.26	\$33,051.09	1	\$2,754.26

4. Construction Disconnected, 14.

IV. Wasted time is costing you millions, so how do you fix it with data management?

People's first instinct when they hear they are spending money on wasted time is to just tell people to stop wasting time. Unfortunately, this is likely to make the problem worse by demoralizing your team. What makes more sense: giving someone 10,000 nails and telling them to hammer as fast as possible, or getting them a nail gun?

If people are going to stop wasting time, they need better tools—an actual solution to their problem. They need a data management tool. What exactly does that mean? Let's break it down.

Data Management Tools in Construction

Construction projects create immense amounts of data: every piece of communication, everything planned to be built, everything actually built—years of activity need to be stored and accessed.

Data management is not something frequently on the mind of construction teams, the current approach is more on-demand, and it often has poor results.

So what is a good data management tool and how do you use it? A good data management platform saves you time by:

- **Automatically organizing and classifying assets**
- **Providing intuitive info retrieval anytime, anywhere**
- **Having a simple user interface requiring virtually no training**
- **Compliments your current info gathering practices so you don't have to change**

Essentially, it eliminates the complications of using digital records by automatically organizing everything your team does.

How are data management tools different to other types of construction software?

Most construction software creates data, and then organizes it within its own system. These are called point solutions. They typically excel at solving one pain point, for instance photo documentation, daily reports, or time tracking.

The problem with point solutions is that they're expensive, and once you get too many of them they start to create a mess of data that can't work together.

Data management tools are the cure for an overabundance of point solutions. They are designed to integrate with all your software and devices to give you a single, organized overview of your projects and everything that's occurring within them.

How would a data management tool eliminate the problem of wasted time? First you need to understand exactly what people are doing when they're not being productive.

5 problems create 93% of unproductive activities ⁵

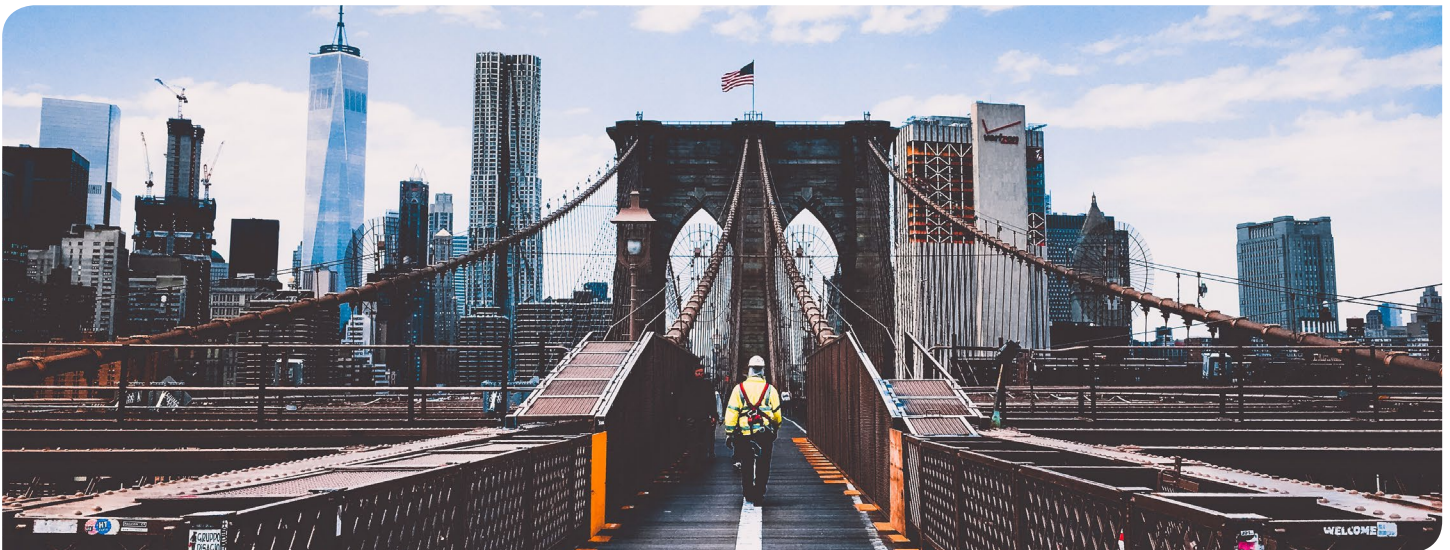
- 1 Poor communication among stakeholders
- 2 Lack of data responsiveness or information delivery
- 3 Confidence in the accuracy of the available data
- 4 Difficulty gathering information
- 5 Inability to collaborate effectively

5. *Construction Disconnected*, 17.

One thing underlies all of these problems: the lack of a central reliable system for people to access and share data.

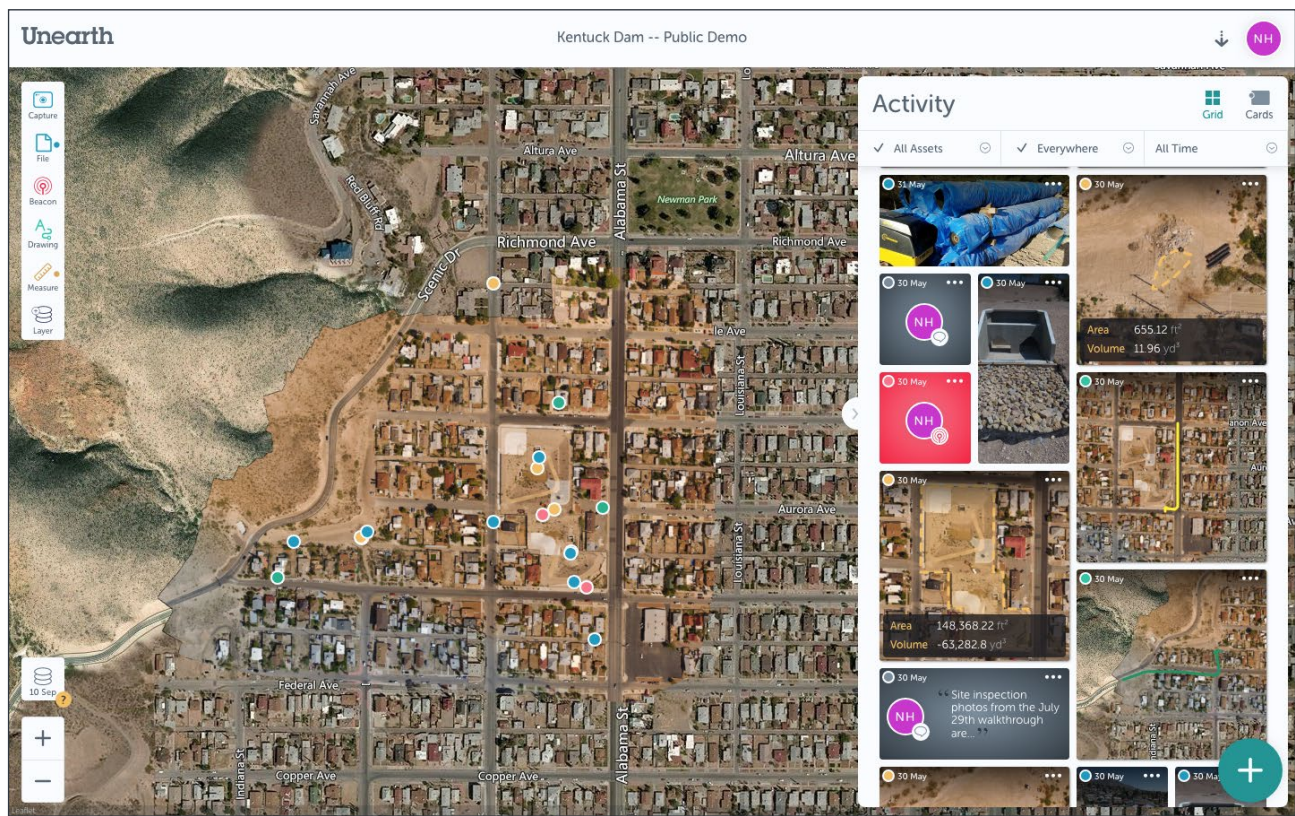
Once you get a data management system in place, you can eliminate the root cause of unproductive work.

Using one system to address a host of problems seems pretty improbable, but realistically the issue is relatively simple. It's easier to demonstrate what we're talking about with some practical examples. The next section will demonstrate how our data management system, **OnePlace™**, solves these problems for our customers.



V. Data Management in Action: Uneath's OnePlace™ software.

Understanding how proper data management prevents companies from losing money on un-productive activities requires context. For this purpose, we'll use our own data management platform, **OnePlace™**, with some practical examples of how the system eliminated this cost for other companies.



What is Uneath's OnePlace™ software?

To be a data management tool, you have to figure out how to not only effectively organize data, but also how to do it automatically. That's why our team landed on a unique system of geo-location. Instead of organizing your data in files and folders, we organize it on a map of your job-site—something that matches the way you actually think about a project. When you need information you can just zoom in to that part of your project, filter by content type, and you'll immediately see all the available data.

Using geolocation is particularly handy because the majority of modern tech attaches GPS data to the files it creates. Which means all we have to do is read that data to ensure it goes to its exact location on your site without you ever having to do anything. Our system essentially ingests all your data, sorts it by location, overlays it on a map of your project, and then gives you the tools you need to analyze and share it.

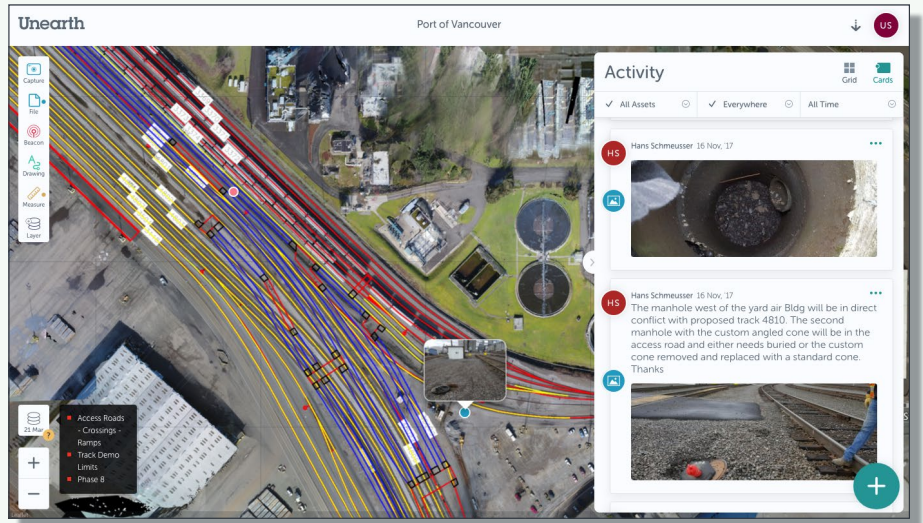
Contractor
Rotschy Inc.

Project
Port of Vancouver Rail Yard

Location
Vancouver, WA

Problems Prevented
Poor stakeholder communication

Summary
Inefficient stakeholder meetings to coordinate an active railyard during construction were causing major delays.



To demonstrate how data management can solve the #1 problem related to wasted time, poor communication among stakeholders, let's take a look at railway project for our customer Rotschy Inc.

Most project stakeholders are not within a convenient distance of one another, or the project. However, everyone still needs to communicate and stay up-to-date, something frequently done in person. This task was especially critical because of the eight different organizations that needed to maintain shipping schedules while the port was under construction.

To try and maintain the ports shipping schedule, project stakeholders and the contractors scheduled weekly in-person meetings. These check-ins not only took hours to complete, they required hours of commuting time because of the distances involved. Even with these meetings, the project was still falling vastly behind schedule because last-minute problems would inevitably arise throughout the week. The GCs clearly needed a system to make commu-

nication instant while providing the contextual data to make it effective.

Using OnePlace™, they overlayed their PDF plans on top of drone imagery of the rail yard. They could then use the platform's annotation tools to mark up the plans and provide real time updates about delays, train schedules, and the impact of the construction.

The new insight was so effective that they took their stakeholder meetings off of the calendar, saving 4-5 hours per week for each of the 30 different stakeholders at 8 different companies. They also managed to eliminate project delays from railyard logistics that were averaging 10-15 hours in backups per week. These were not only setting the construction schedule behind, they were causing expensive delays on shipments from the port.

Just by getting a handle on their data organization and giving people access to the information they needed, this port project was able to streamline operations to the tune of almost \$15,000 in cost savings per week.

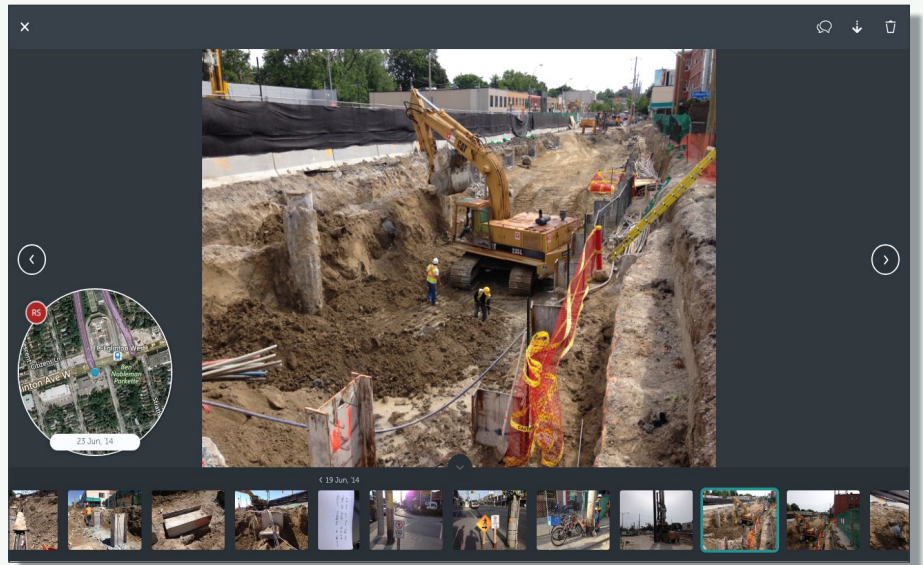
Contractor
Obayashi

Project
Eglinton Square Tunnel

Location
Toronto, Canada

Problems Prevented
Difficulty gathering information, lack of data responsiveness, unconfident in data accuracy

Summary
After completing a project, Obayashi ran into a dispute and needed to sort through 50k+ photos for evidence.



When teams can't get relevant information from available data, or even worse, if they can't find what they need at all, complication-focused work immediately begins. Fortunately, with the right data management tools, these problems can be resolved—even after a project has been completed.

The general contractor, Obayashi, had recently completed a tunnel project that ran into a dispute. The dispute went to court—and they were required to prove that construction met certain progress milestones within specific time frames. The team had taken 50k+ photos to document the project, but no one

had made sure they were organized—meaning they likely had the evidence they needed, but no way to actually uncover it.

By using Uneath for data management, they were able to bulk upload all 50k+ photos and have the system parse the associated metadata to automatically sort them according to location and date on a map of their jobsite. This provided their team and their lawyers with a simple way to immediately understand what work had been completed where and when—saving hundreds of hours and hundreds of thousands of dollars in spending on complication-focused work.

The previous two examples illustrate how quickly construction data can get out of control, but also how easily the problem can be remedied. Data management doesn't have to be a complex process or a massive undertaking. With a few simple steps the industry can reverse billions of dollars in unnecessary spending.

VI. How can you turn this white paper into savings for your company?

It's abundantly clear that almost every construction firm has a significant amount of money to be saved by adopting data management tools. Here's a practical guide to getting started:

1 Get a realistic assessment of how much money is on the table.

Use the table we outlined in section III to get an estimate of how much money the average firm of your size is losing to this problem. Use your knowledge of

your company's best practices to get an idea of whether you're going to be significantly over or under the national average.

2 Take this number to the people who matter.

Bring this problem to light and start a discussion with your team about why it matters. Get people fired up

about how easy of a way this is to save significant amounts of cash.

3 Weigh your options.

The data management industry is still relatively new for construction, and the options are somewhat slim. Unearth is one of the first players focused entirely on

organizing your construction data and improving your data management workflows. Search around for some other options and then compare your top 3 choices.

4 Reach out to your favorite data management softwares.

Part of the job of everyone in the data management space is to do most of the heavy lifting for their customers. Everyone you talk to should be able to roll

out a detailed program for you that will help you get your team up and running as soon as possible with minimal effort.

5 Get moving on a pilot program.

Once you've heard from all the companies you're considering, get started on a pilot program. This is a short-term subscription at a discounted cost that should help you become familiar with the software

and make sure everything is working as expected. At the end of this term, you should have the option to continue or terminate your subscription with no penalties.

6 Be ready to keep pushing forward.

Remain committed and focus on expanding your capabilities, because the benefits of digital innovation increase exponentially with scale. Getting one

project in order will show you a small ROI. Getting your entire company on board will revolutionize your business.



Have questions about any aspect of this white paper, or just curious to learn more about data management in construction? Reach out to us at info@unearthlabs.com or call us at (206) 257-2682.

Be sure to check out our blog, unearthlabs.com/punchlist, for even more construction tech news and information.